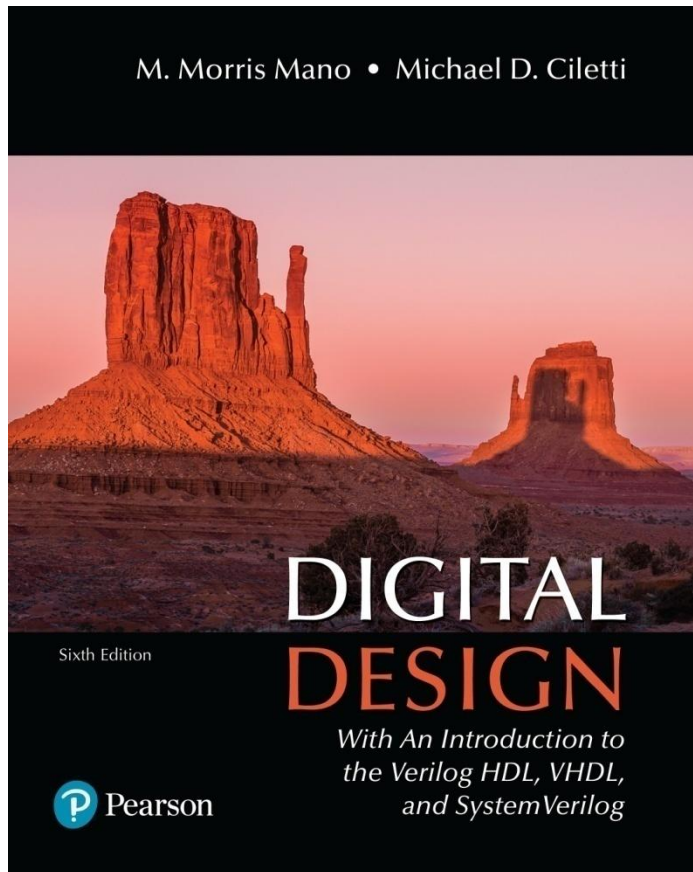


Digital Design

With an Introduction to the Verilog HDL, VHDL, and SystemVerilog

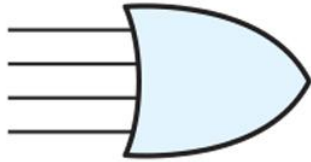
6th Edition



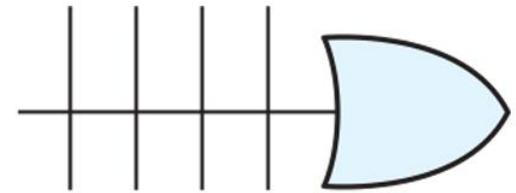
Chapter 07

Memory and Programmable Logic

Figure 7.1
Conventional and array logic diagrams for OR gate.



(a) Conventional symbol



(b) Array logic symbol

Figure 7.2
Block diagram of a memory unit.

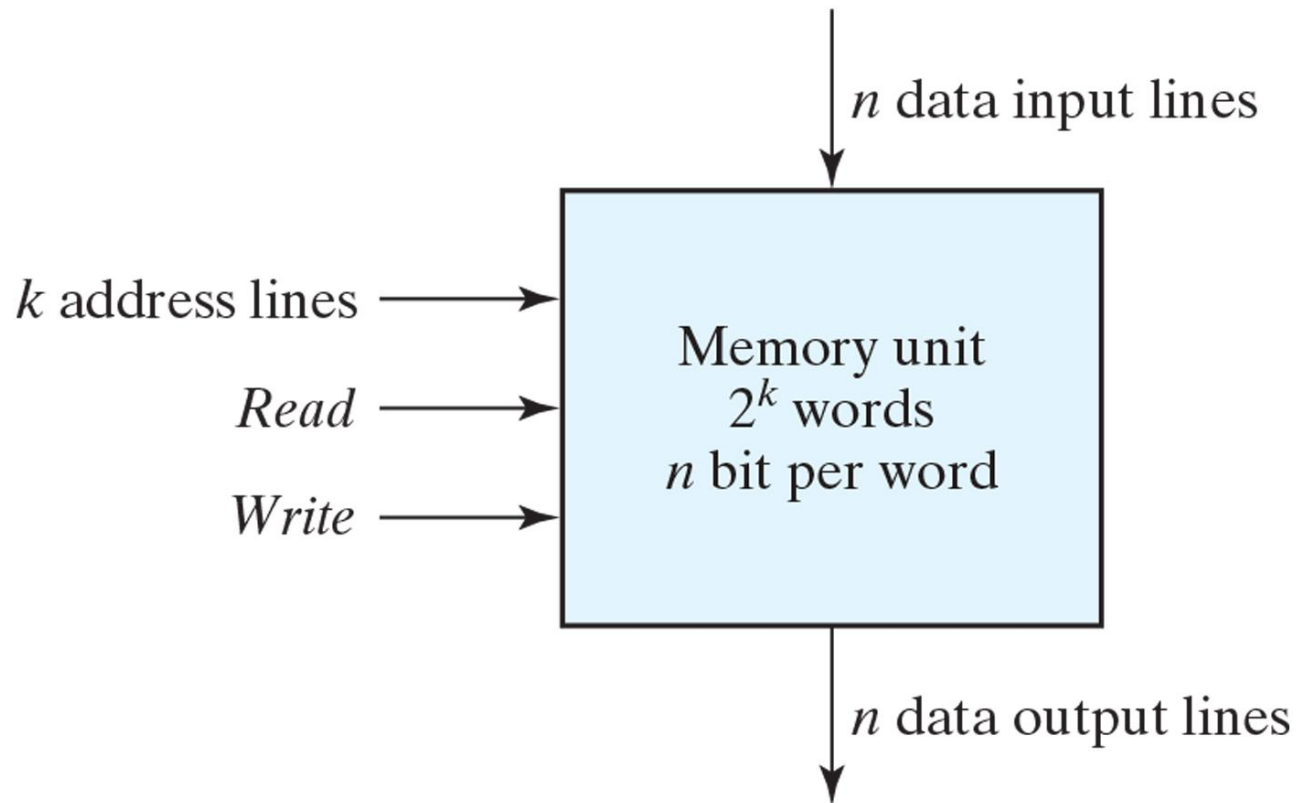


Figure 7.3
Contents of a 1024 × 16 memory.

Memory address		Memory content
Binary	Decimal	
0000000000	0	1011010101011101
0000000001	1	1010101110001001
0000000010	2	0000110101000110
	⋮	⋮
1111111101	1021	1001110100010100
1111111110	1022	0000110100011110
1111111111	1023	1101111000100101

Table 7.1
Control Inputs to Memory Chip.

Memory Enable	Read/Write	Memory Operation
0	X	None
1	0	Write to selected word
1	1	Read from selected word

Figure 7.4

Memory cycle timing waveforms.

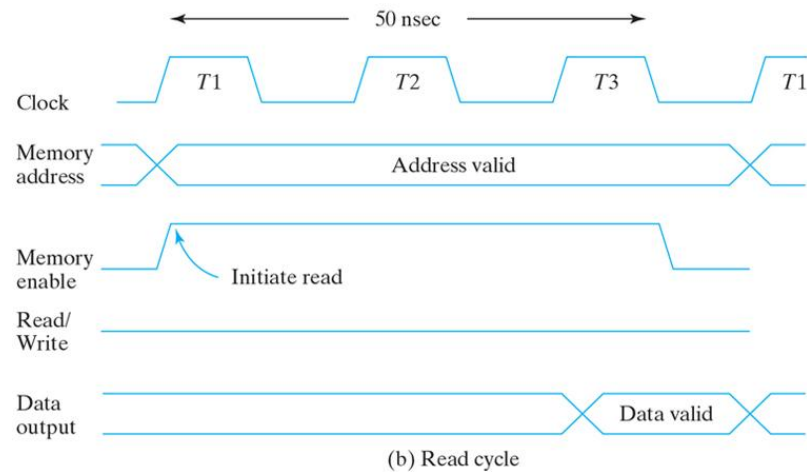
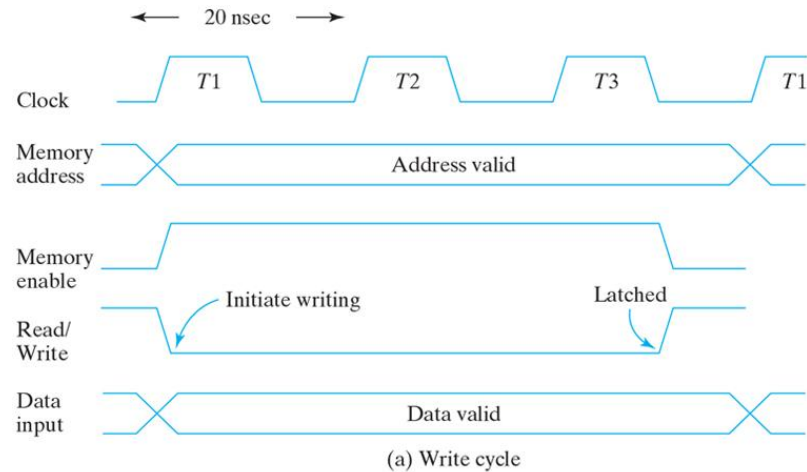


Figure 7.5
Memory cell.

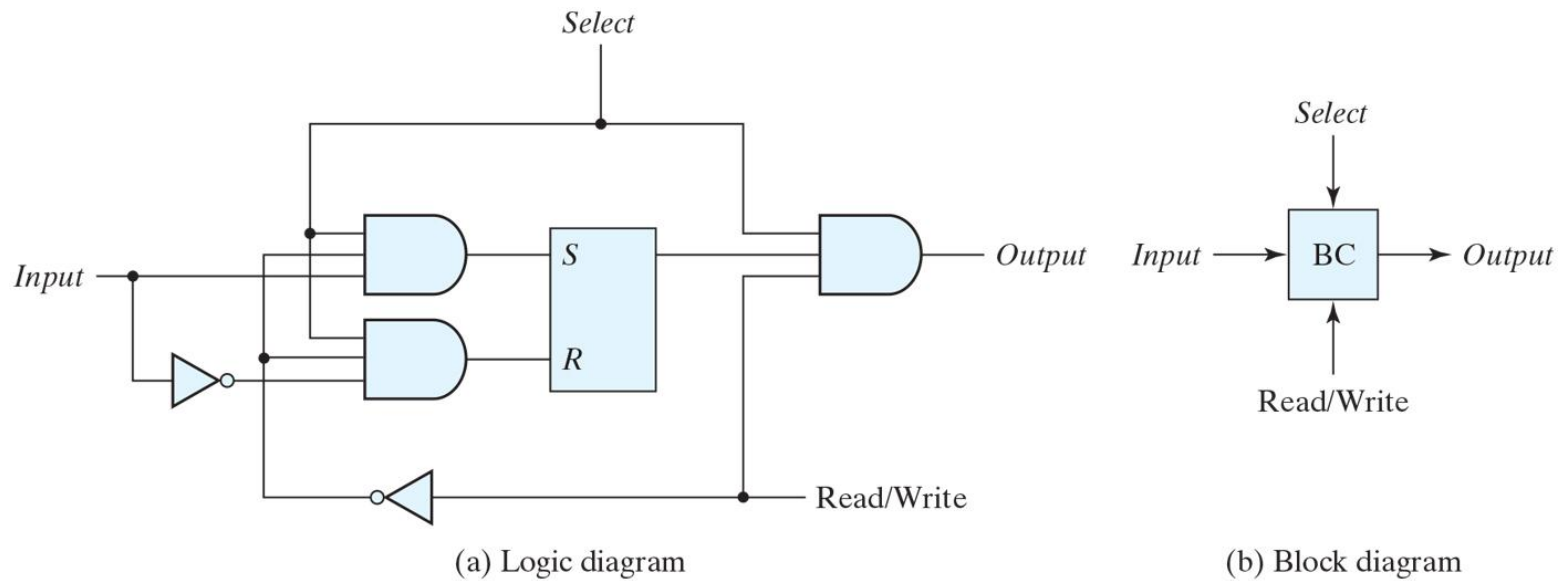


Figure 7.6
Diagram of a 4 × 4 RAM.

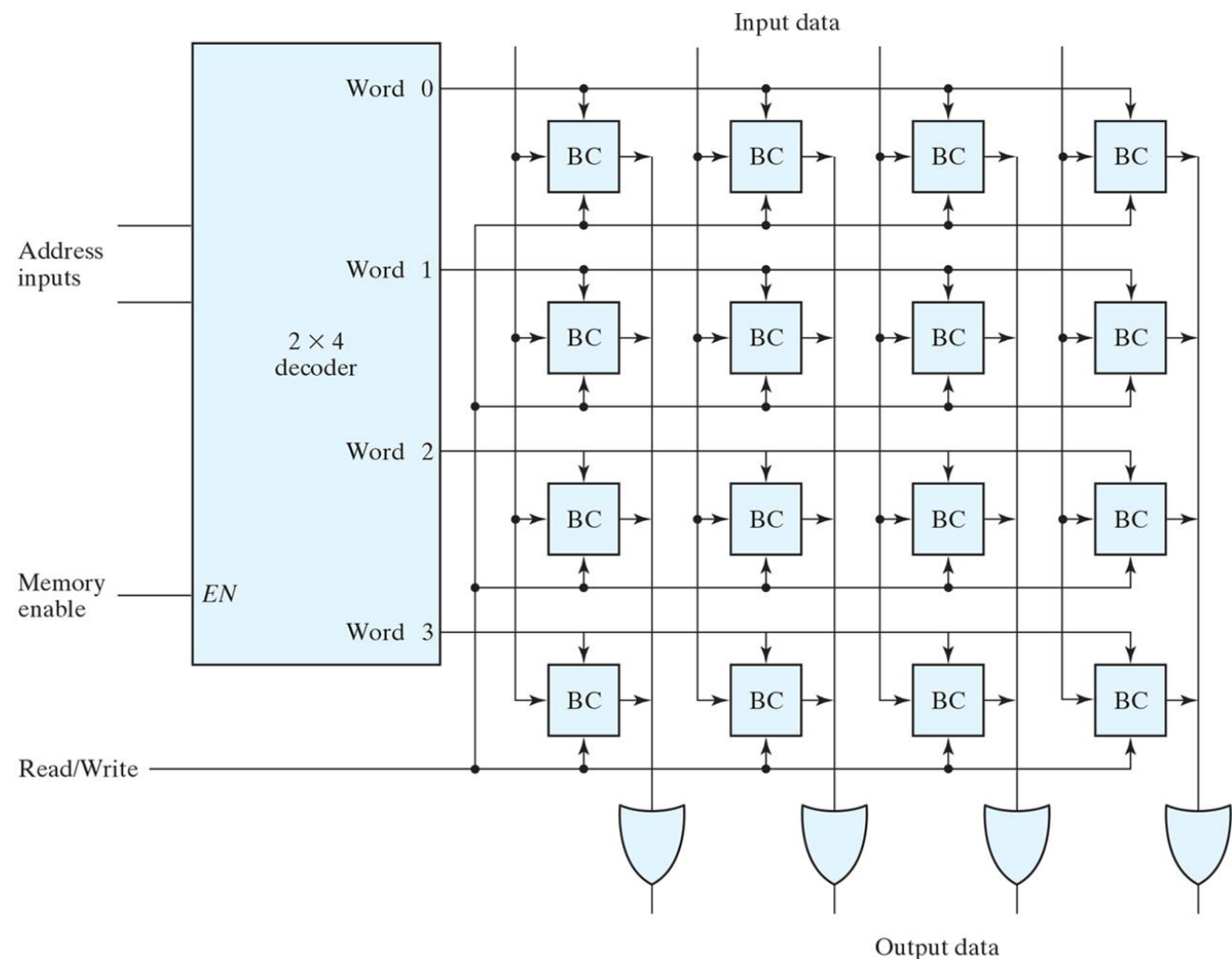


Figure 7.7
Two-dimensional decoding structure for a 1K-word memory.

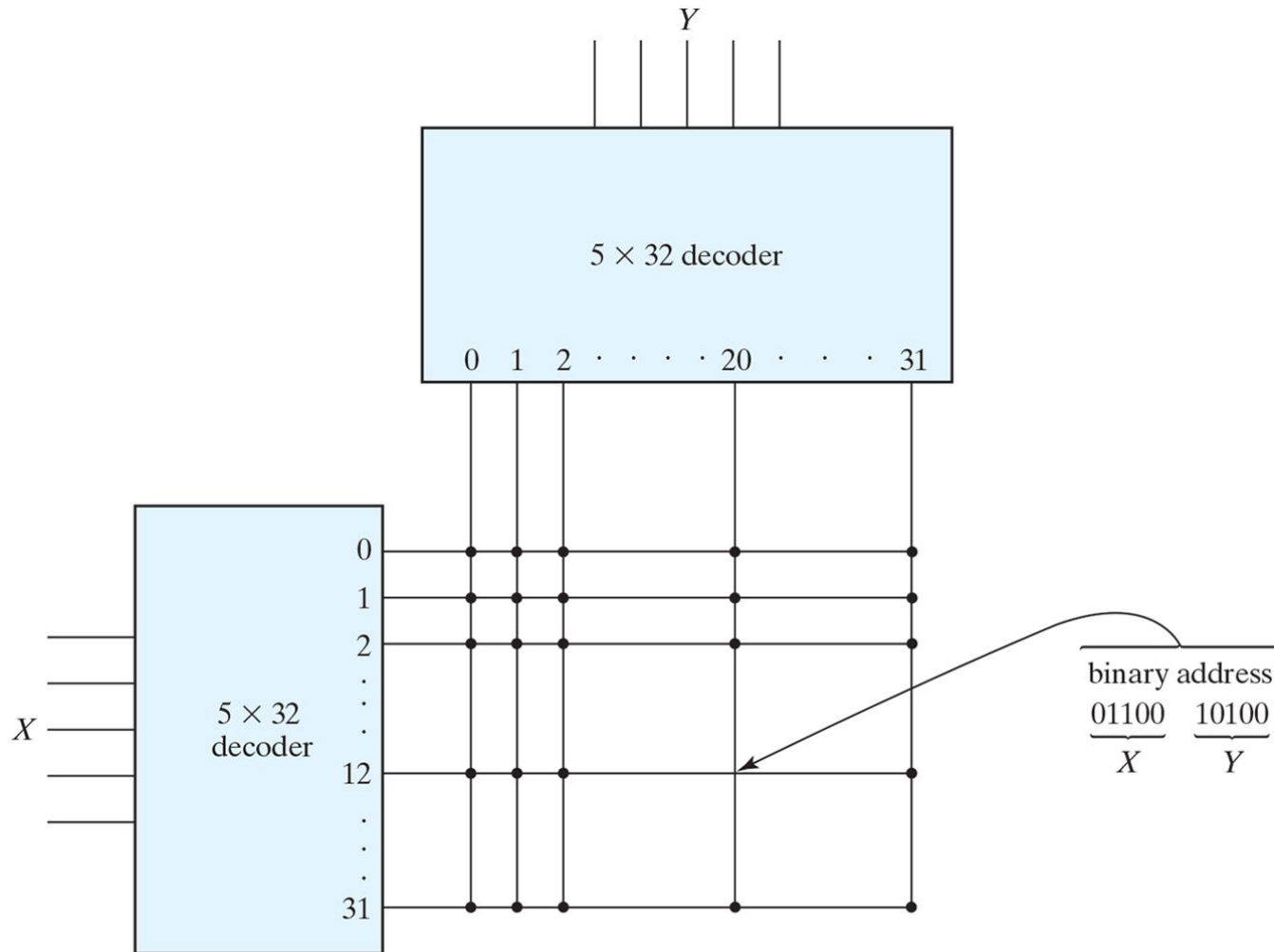


Figure 7.8
Address multiplexing for a 64K DRAM.

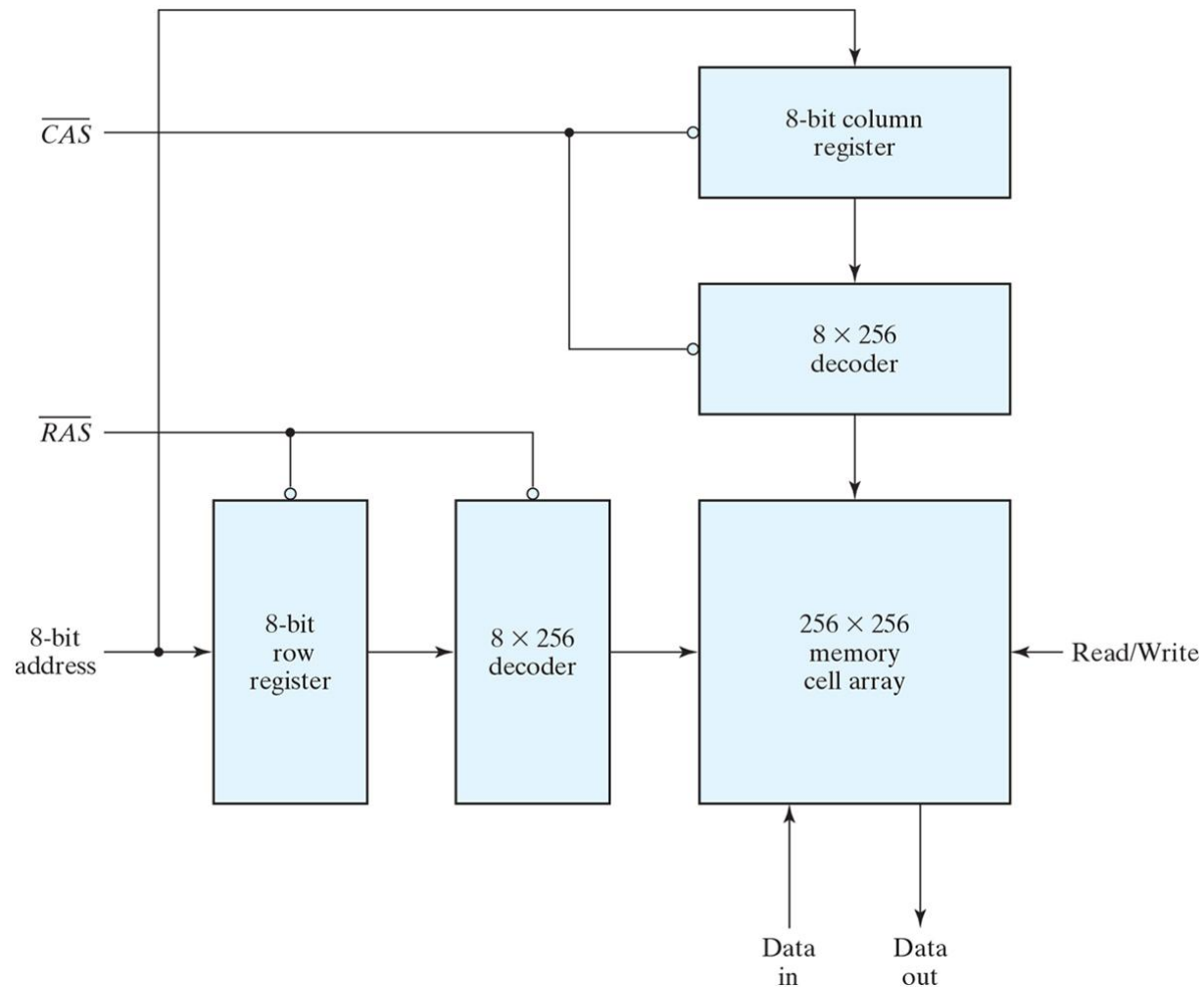


Table 7.2
Range of Data Bits for k Check Bits.

Number of Check Bits, k	Range of Data Bits, n
3	2–4
4	5–11
5	12–26
6	27–57
7	58–120

Figure 7.9
ROM block diagram.

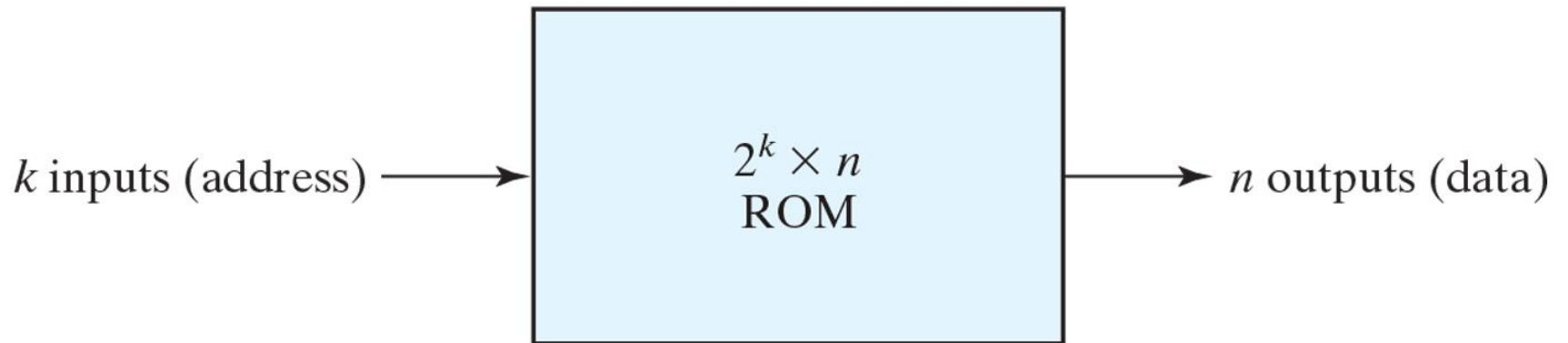


Figure 7.10
Internal logic of a 32×8 ROM.

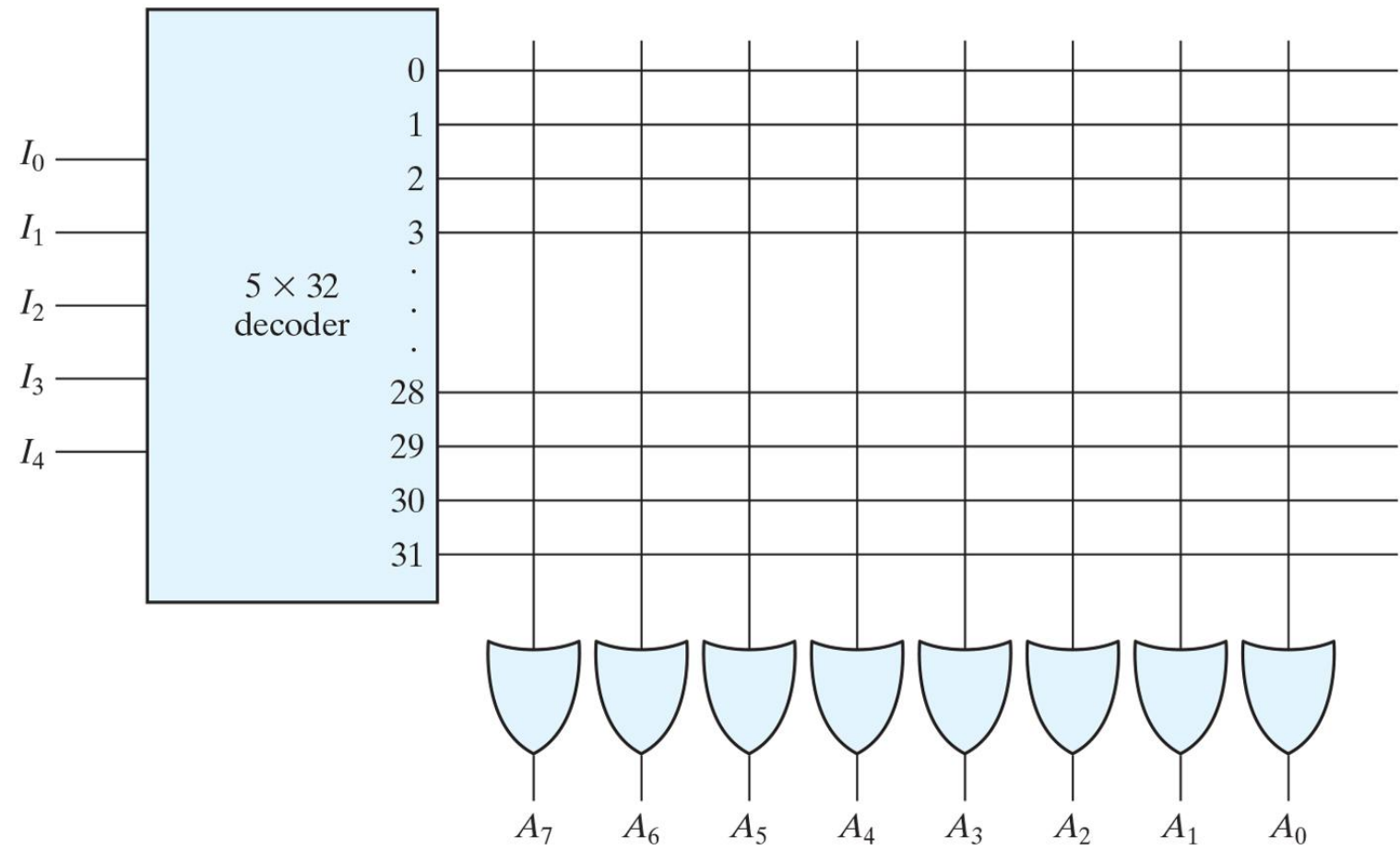


Table 7.3
ROM Truth Table (Partial).

Inputs					Outputs							
I_4	I_3	I_2	I_1	I_0	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0
0	0	0	0	0	1	0	1	1	0	1	1	0
0	0	0	0	1	0	0	0	1	1	1	0	1
0	0	0	1	0	1	1	0	0	0	1	0	1
0	0	0	1	1	1	0	1	1	0	0	1	0
		$\vdots$							$\vdots$			
1	1	1	0	0	0	0	0	0	1	0	0	1
1	1	1	0	1	1	1	1	0	0	0	1	0
1	1	1	1	0	0	1	0	0	1	0	1	0
1	1	1	1	1	0	0	1	1	0	0	1	1

Figure 7.11
Programming the ROM according to Table 7.3.

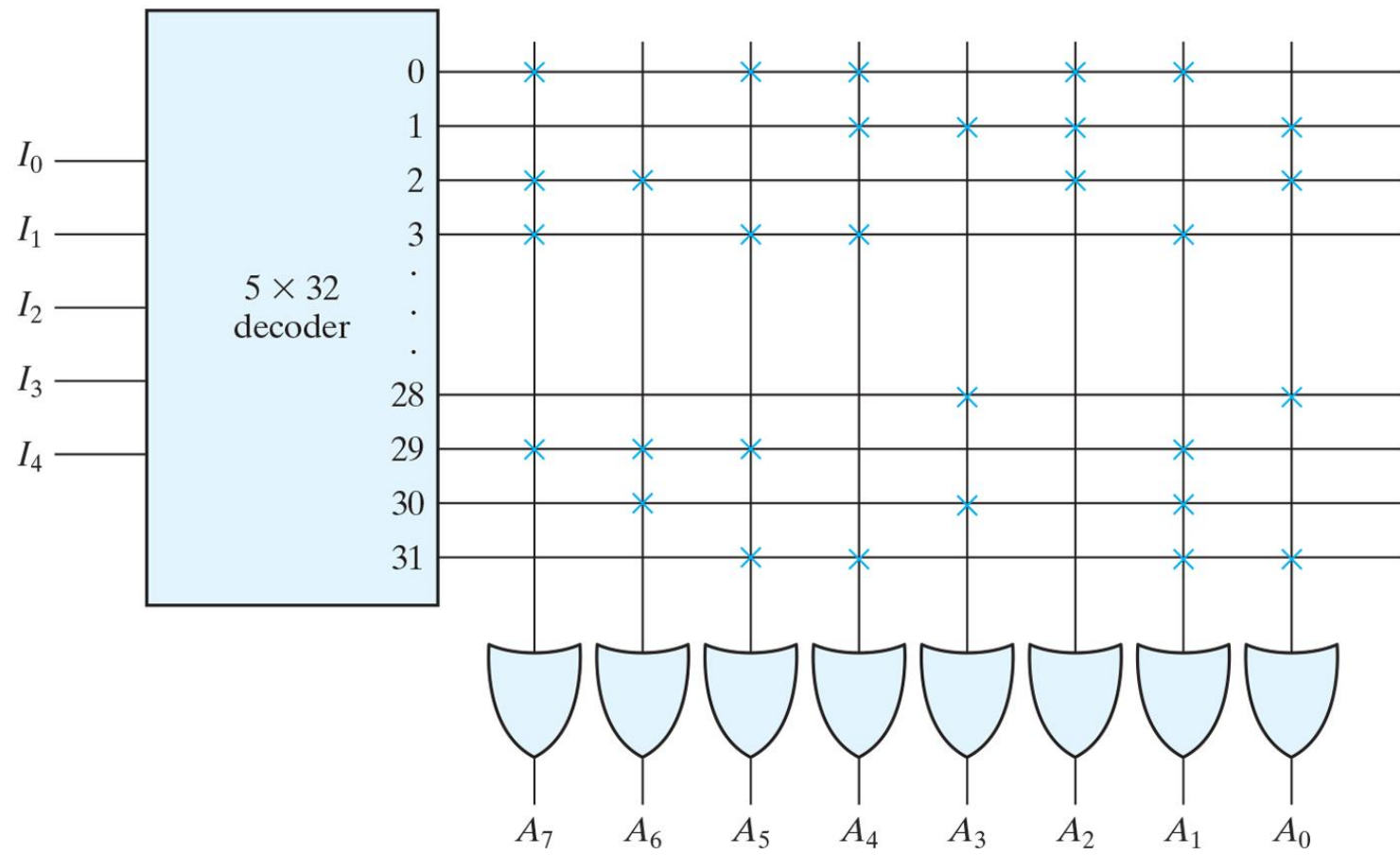
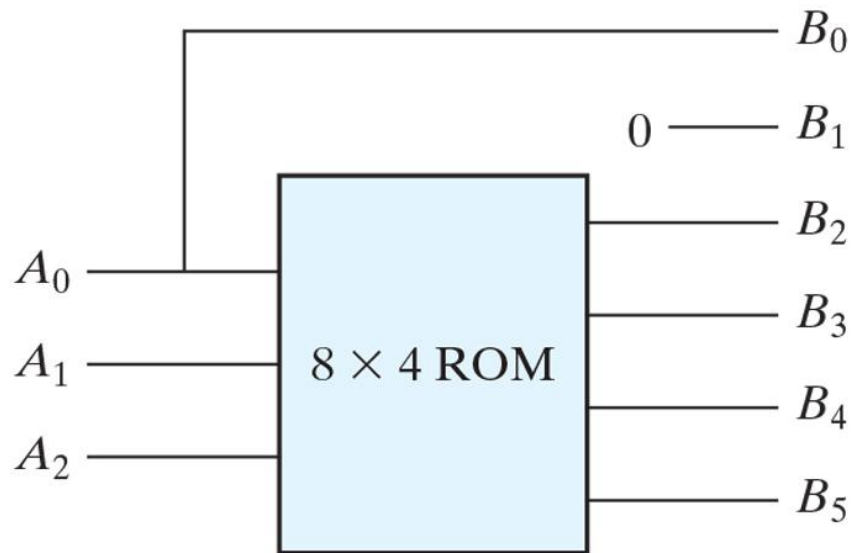


Table 7.4
Truth Table for Circuit of Example 7.1.

Inputs			Outputs						Decimal
A_2	A_1	A_0	B_5	B_4	B_3	B_2	B_1	B_0	
0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	1	1
0	1	0	0	0	0	1	0	0	4
0	1	1	0	0	1	0	0	1	9
1	0	0	0	1	0	0	0	0	16
1	0	1	0	1	1	0	0	1	25
1	1	0	1	0	0	1	0	0	36
1	1	1	1	1	0	0	0	1	49

Figure 7.12
ROM implementation of Example 7.1.



(a) Block diagram

A_2	A_1	A_0	B_5	B_4	B_3	B_2
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	1
0	1	1	0	0	1	0
1	0	0	0	1	0	0
1	0	1	0	1	1	0
1	1	0	1	0	0	1
1	1	1	1	1	0	0

(b) ROM truth table

Figure 7.13
Basic configuration of three PLDs.



(a) Programmable read-only memory (PROM)



(b) Programmable array logic (PAL)



(c) Programmable logic array (PLA)

Figure 7.14
PLA with three inputs, four product terms, and two outputs.

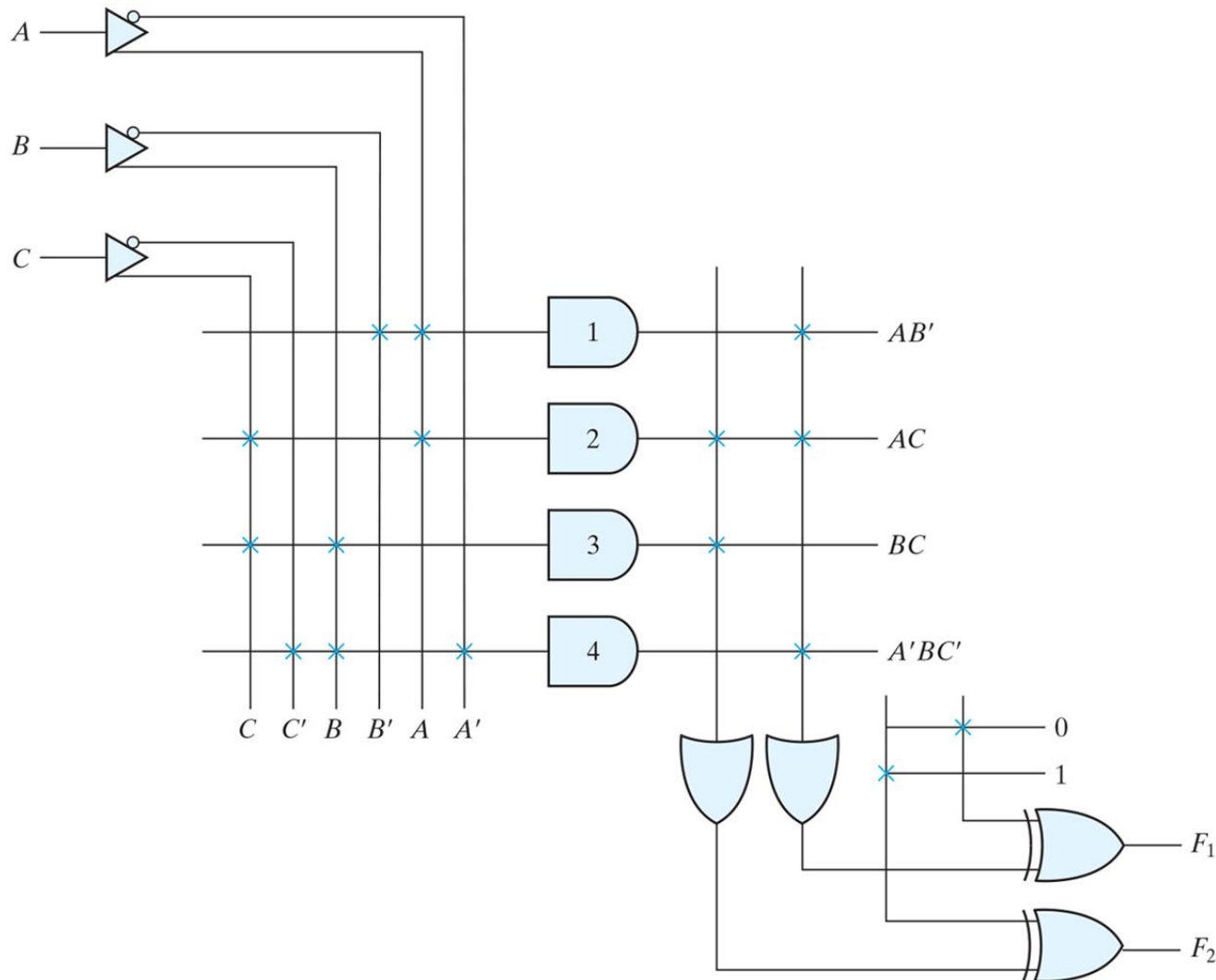


Table 7.5
PLA Programming Table.

		Inputs			Outputs (T) (C)	
	Product Term	<i>A</i>	<i>B</i>	<i>C</i>	<i>F</i> ₁	<i>F</i> ₂
<i>AB'</i>	1	1	0	—	1	—
<i>AC</i>	2	1	—	1	1	1
<i>BC</i>	3	—	1	1	—	1
<i>A'BC'</i>	4	0	1	0	1	—

Note: See text for meanings of dashes.

Figure 7.15
Solution to Example 7.2.

PLA programming table						
		Inputs			Outputs	
Product term		A	B	C	(C) F_1	(T) F_2
AB	1	1	1	–	1	1
AC	2	1	–	1	1	1
BC	3	–	1	1	1	–
$A'B'C'$	4	0	0	0	–	1

		B			
A	BC	00	01	11	10
	0	m_0 1	m_1 1	m_3 0	m_2 1
1	1	m_4 1	m_5 0	m_7 0	m_6 0

		B			
A	BC	00	01	11	10
	0	m_0 1	m_1 0	m_3 0	m_2 0
1	1	m_4 0	m_5 1	m_7 1	m_6 1

Figure 7.16
PAL with four inputs, four outputs, and a three-wide AND–OR structure.

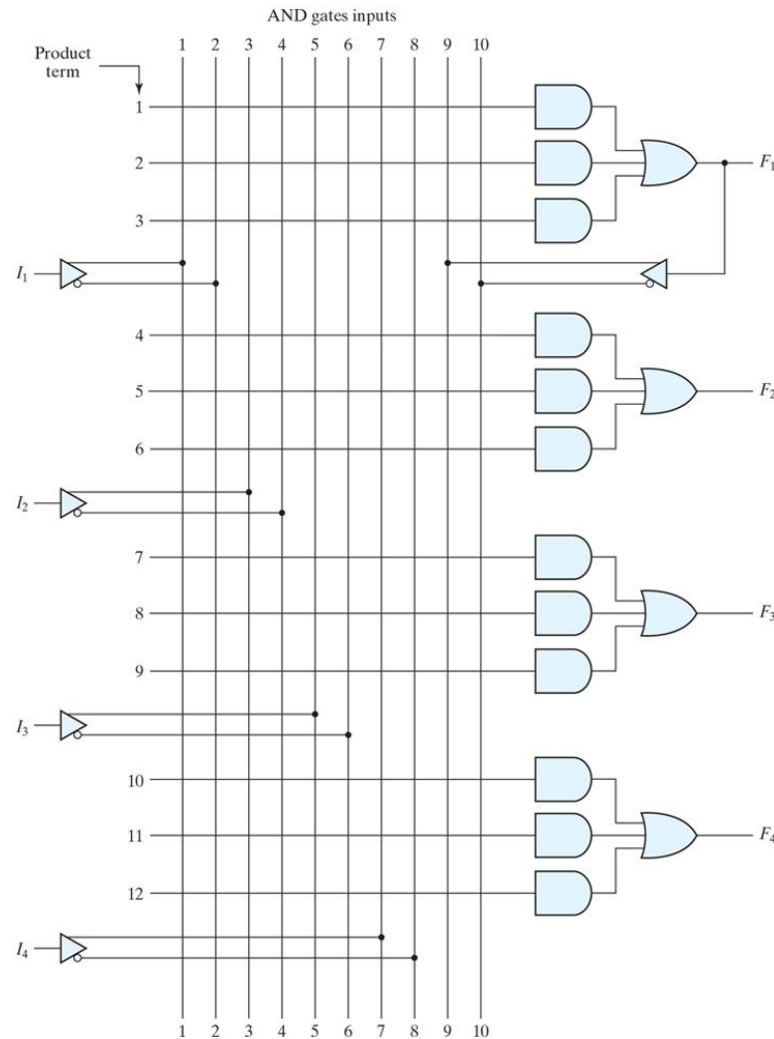


Table 7.6
PAL Programming Table.

Product Term	AND Inputs				<i>w</i>	Outputs
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>		
1	1	1	0	—	—	$w = ABC' + A'B'CD'$
2	0	0	1	0	—	
3	—	—	—	—	—	
4	1	—	—	—	—	$x = A + BCD$
5	—	1	1	1	—	
6	—	—	—	—	—	
7	0	1	—	—	—	$y = A'B + CD + B'D'$
8	—	—	1	1	—	
9	—	0	—	0	—	
10	—	—	—	—	1	$z = w + AC'D' + A'B'C'D$
11	1	—	0	0	—	
12	0	0	0	1	—	

Figure 7.17

Fuse map for PA L as specified in Table 7.6.

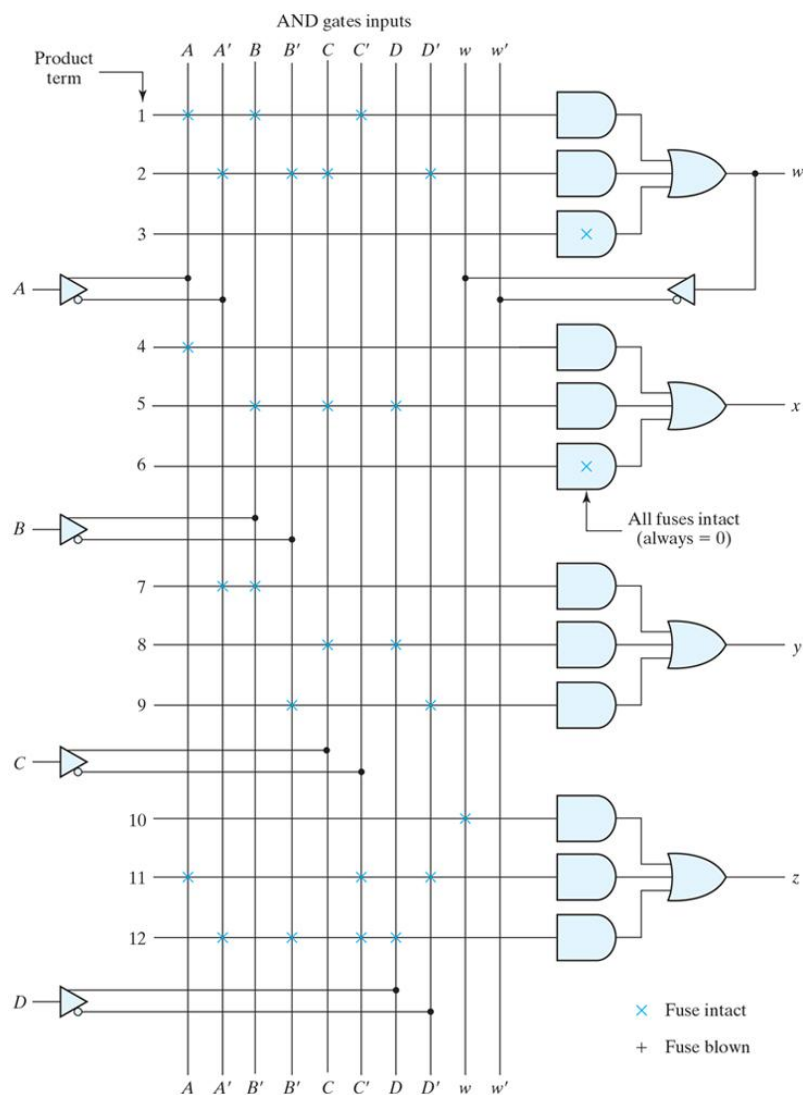


Figure 7.18
Sequential programmable logic device.

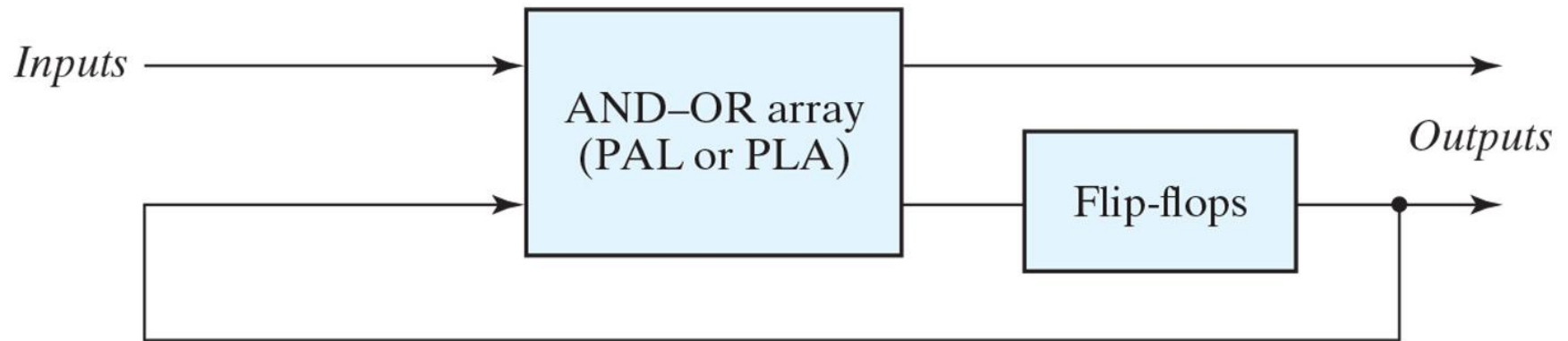


Figure 7.19
Basic macrocell logic.

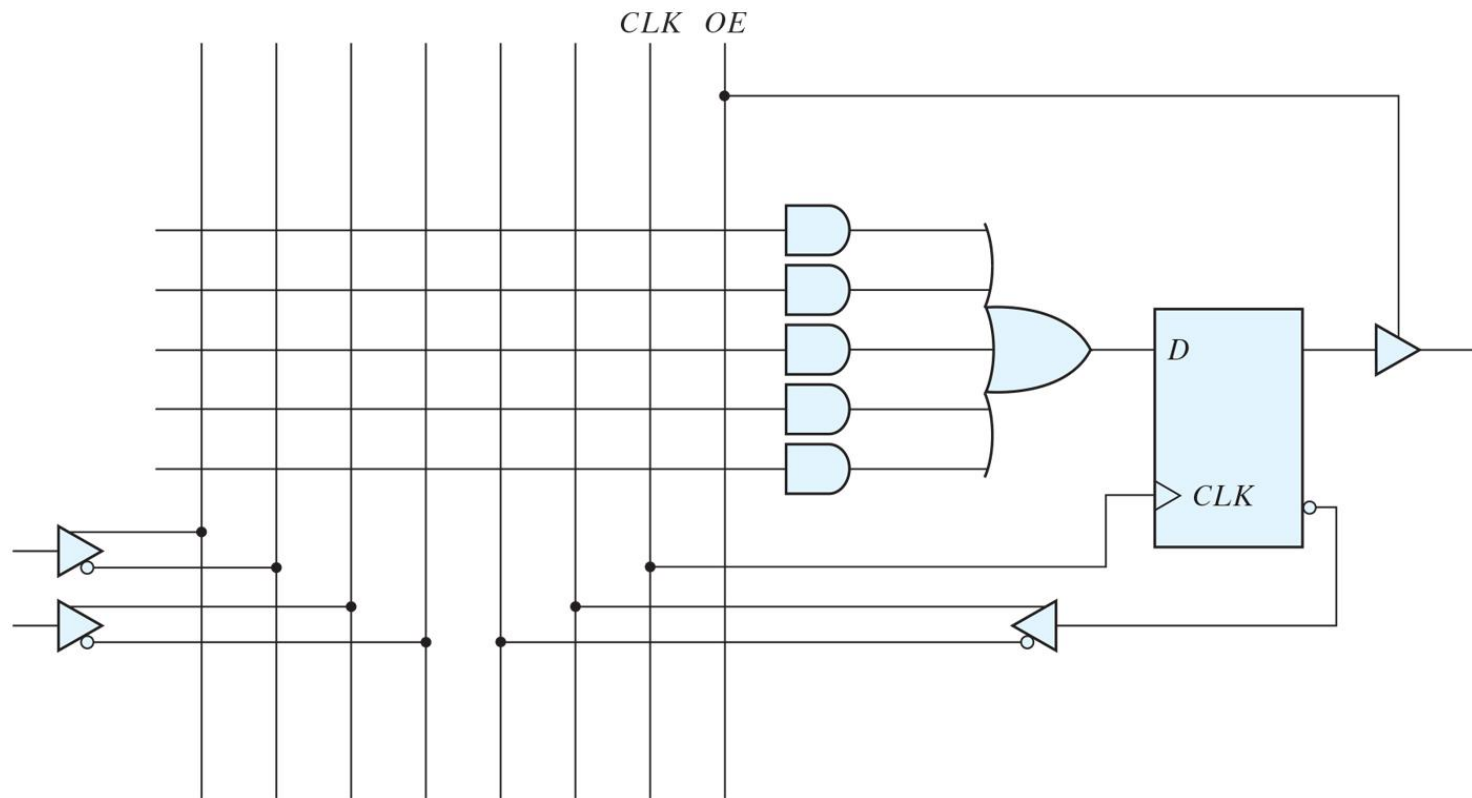


Figure 7.20
General CPLD configuration.

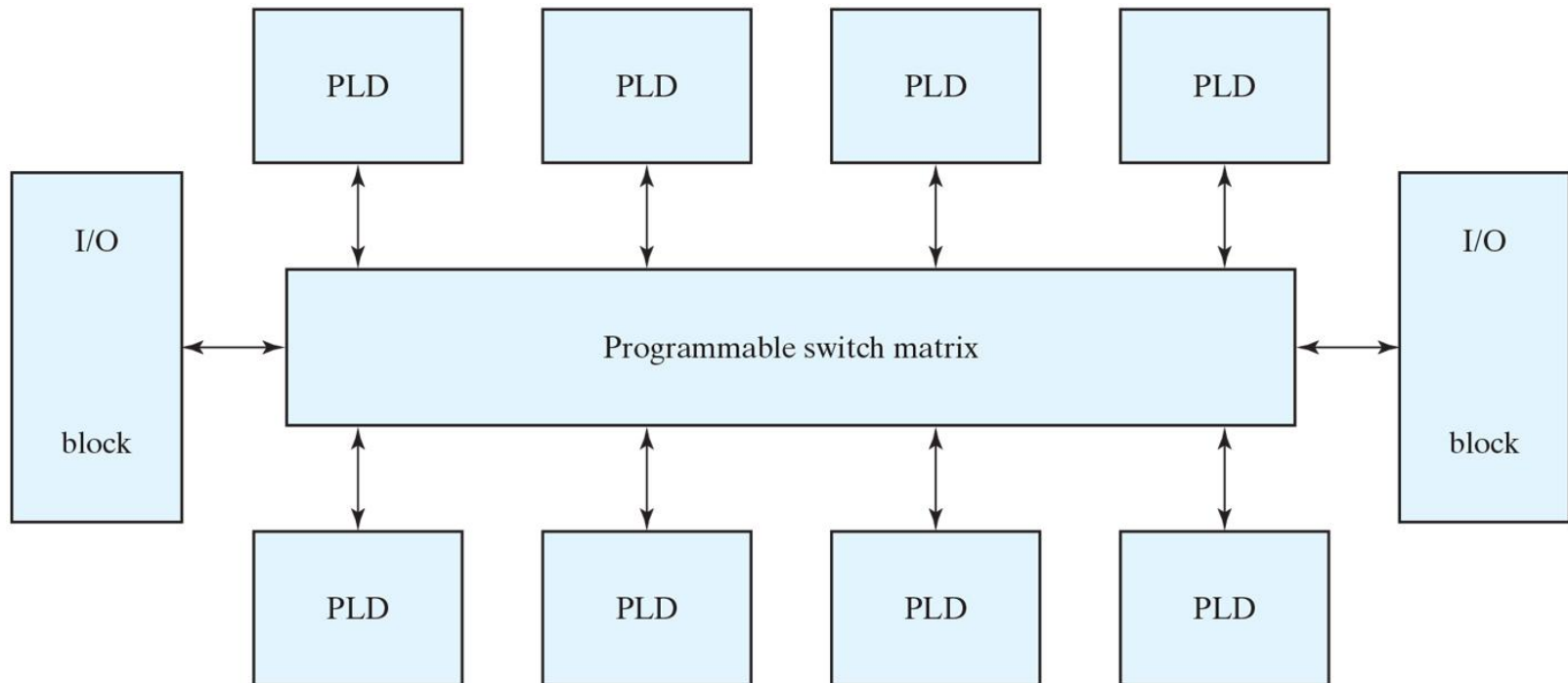


Figure 7.21
Basic architecture of Xilinx Spartan and predecessor devices.

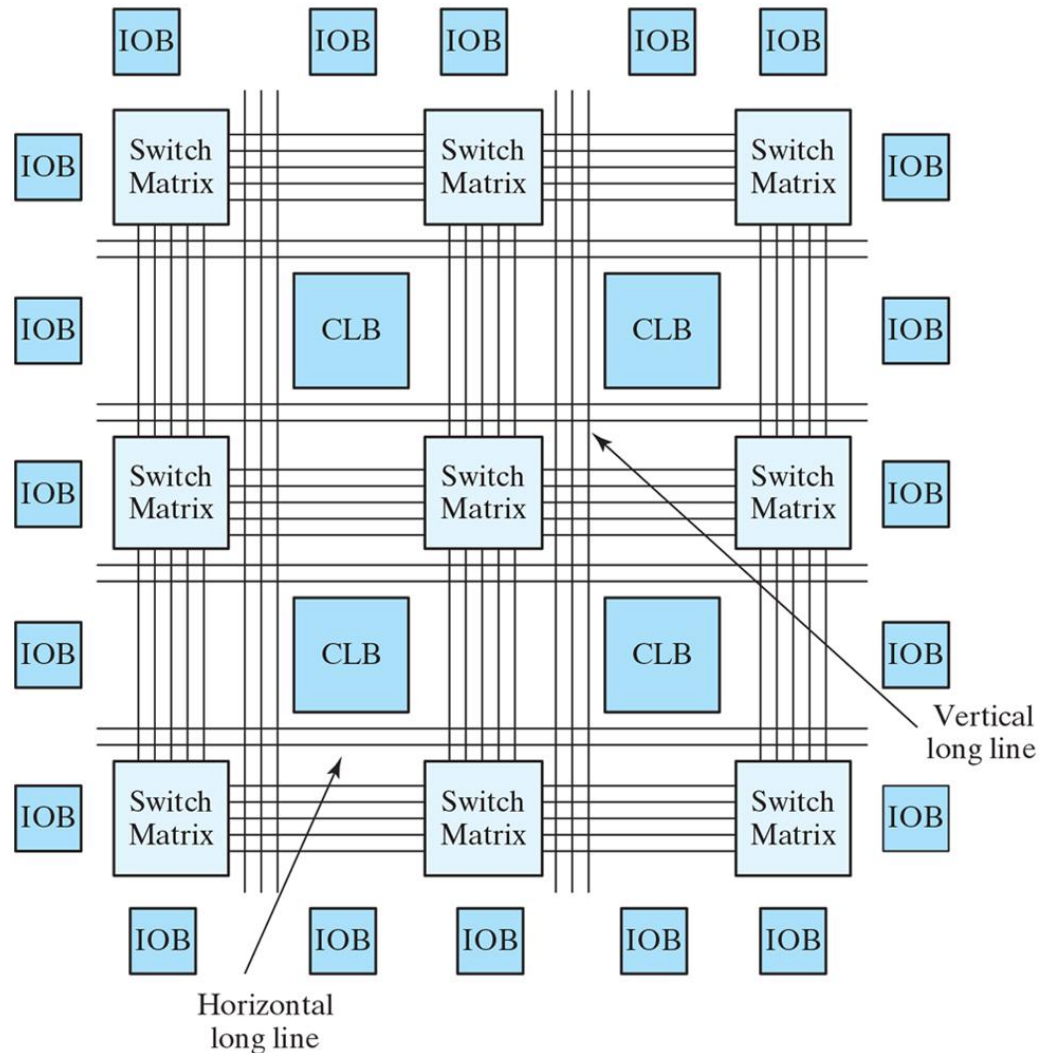


Figure 7.22
CLB architecture.

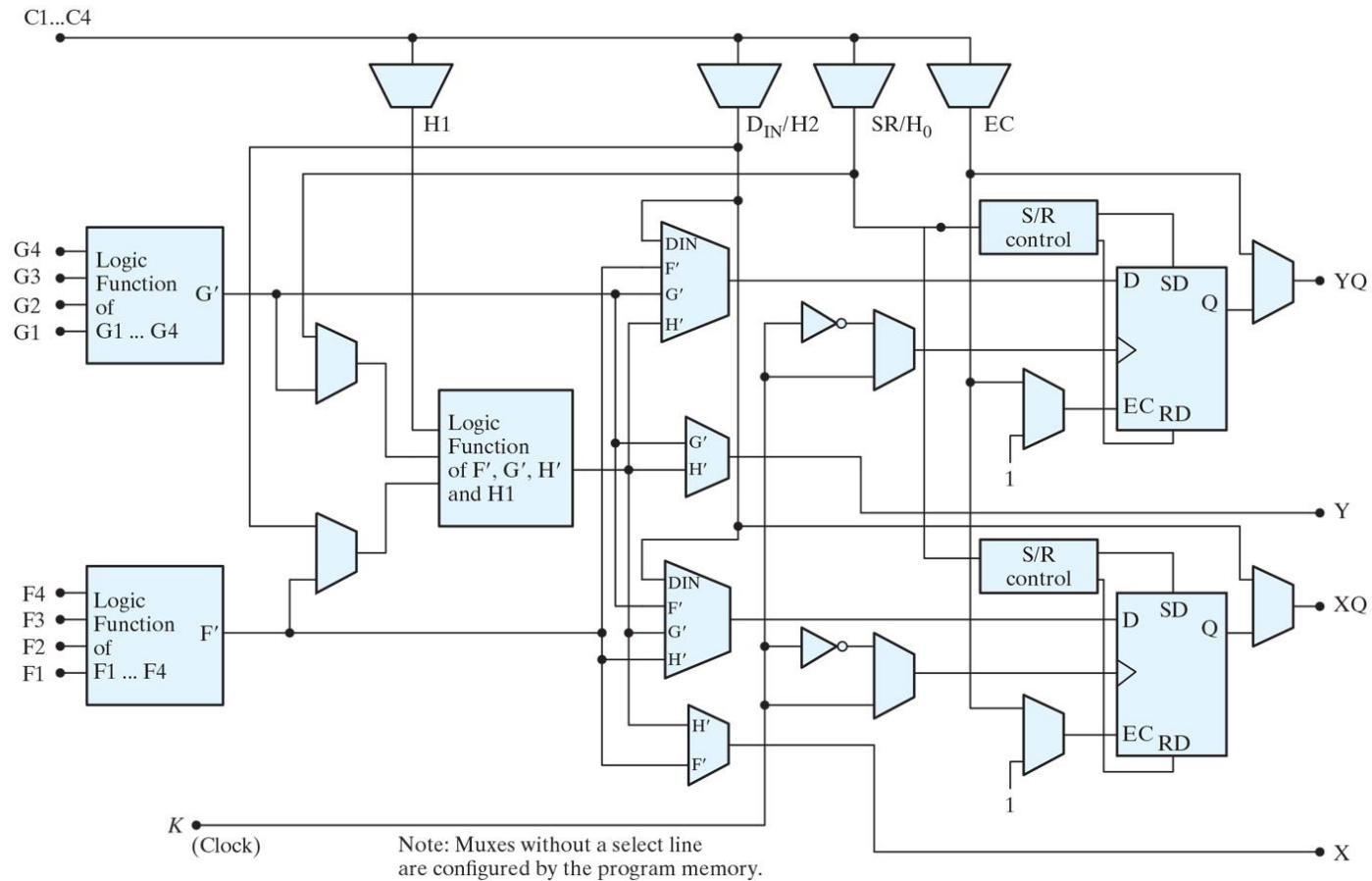


Figure 7.23
RAM cell controlling a PIP transmission gate.

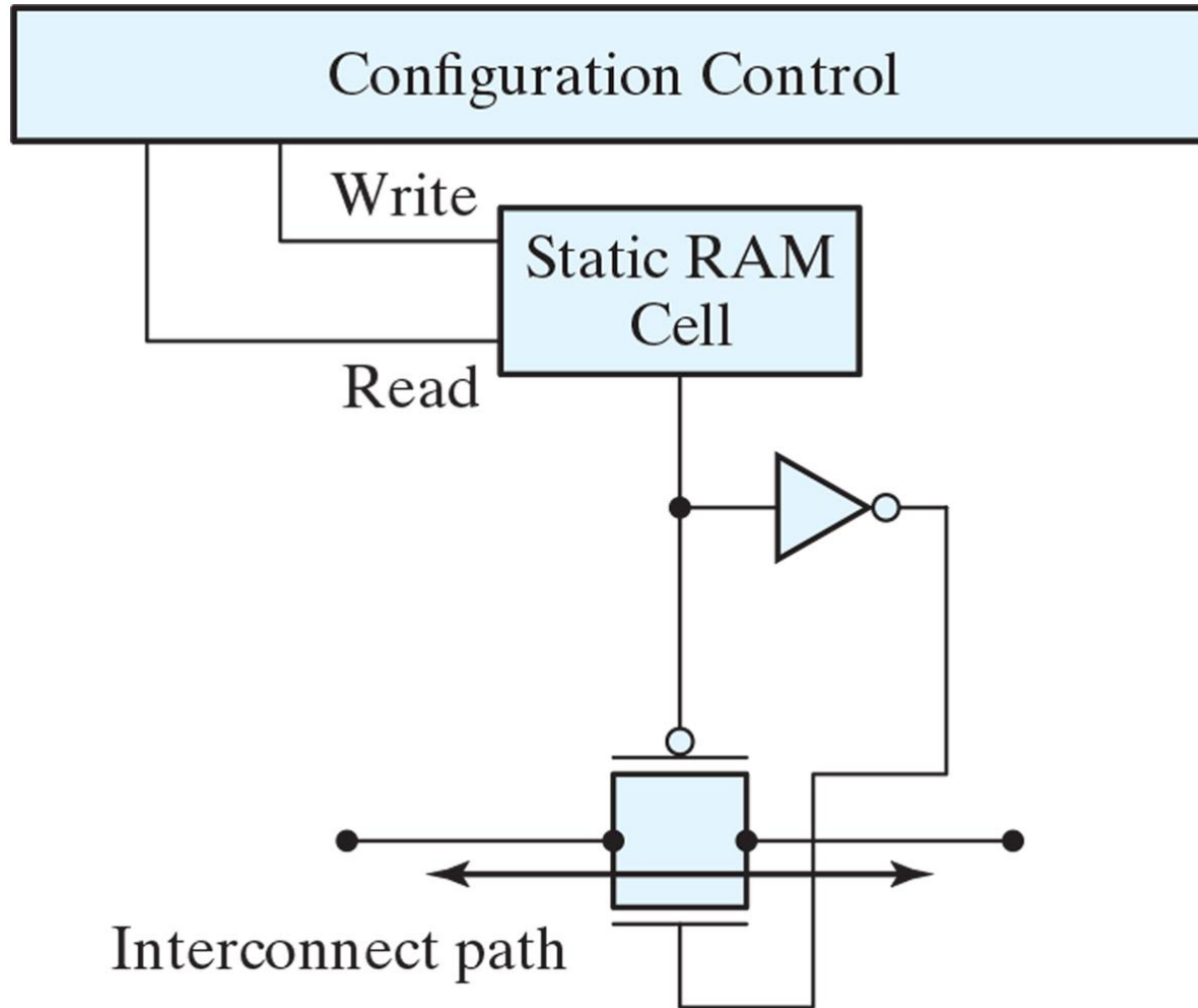


Figure 7.24
Circuit for a programmable PIP.

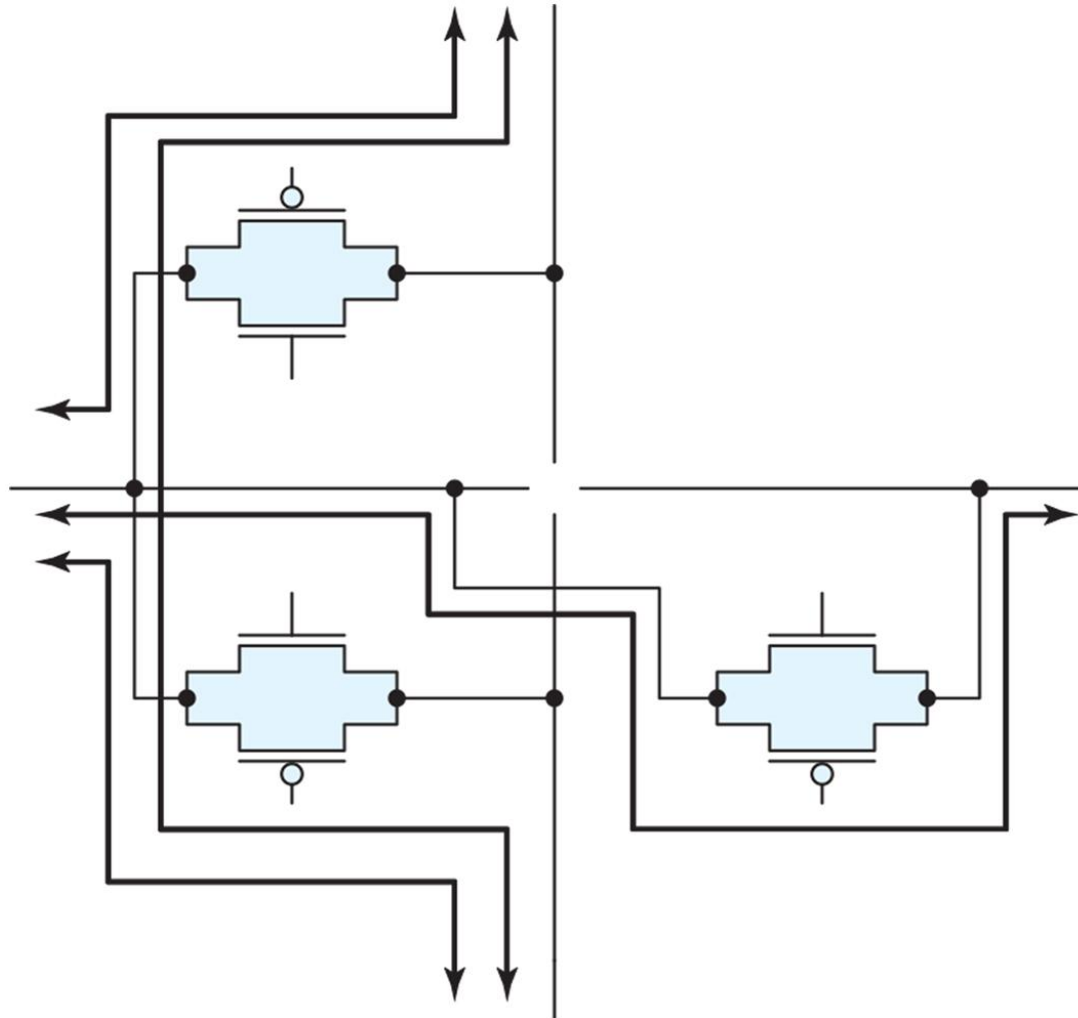


Figure 7.25
XC4000 series IOB.

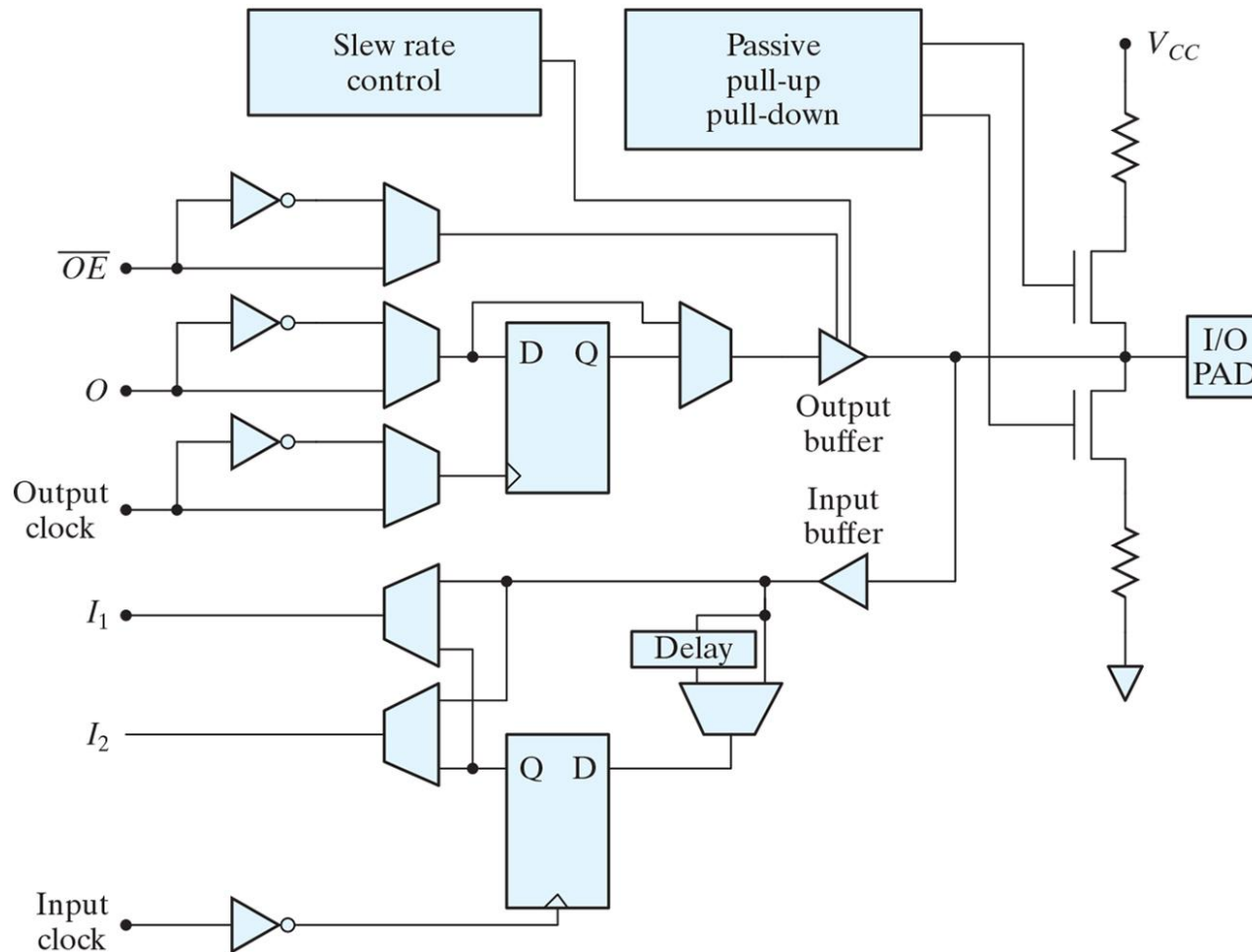


Figure 7.26
Distributed RAM cell formed from a lookup table.

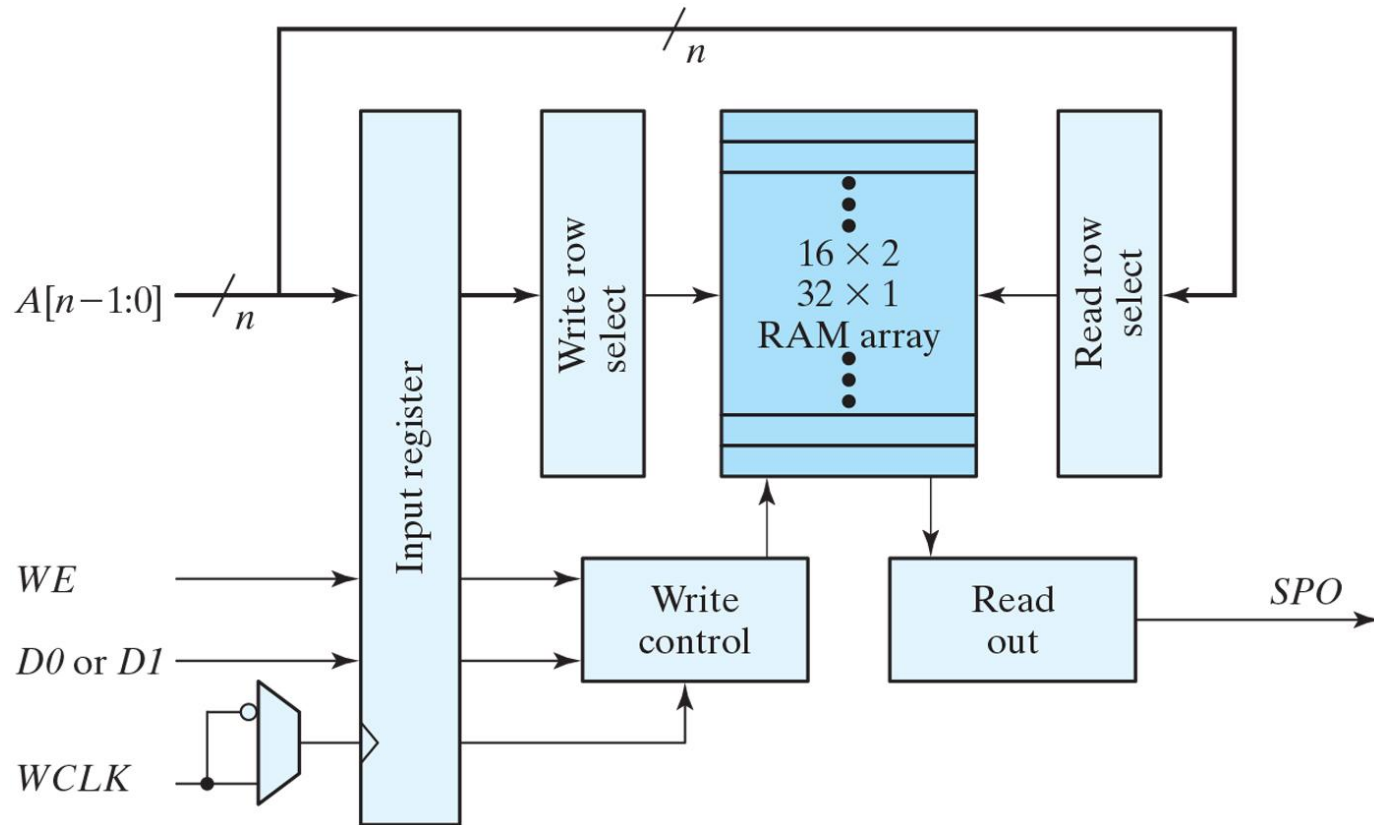


Figure 7.27
Spartan dual-port RAM.

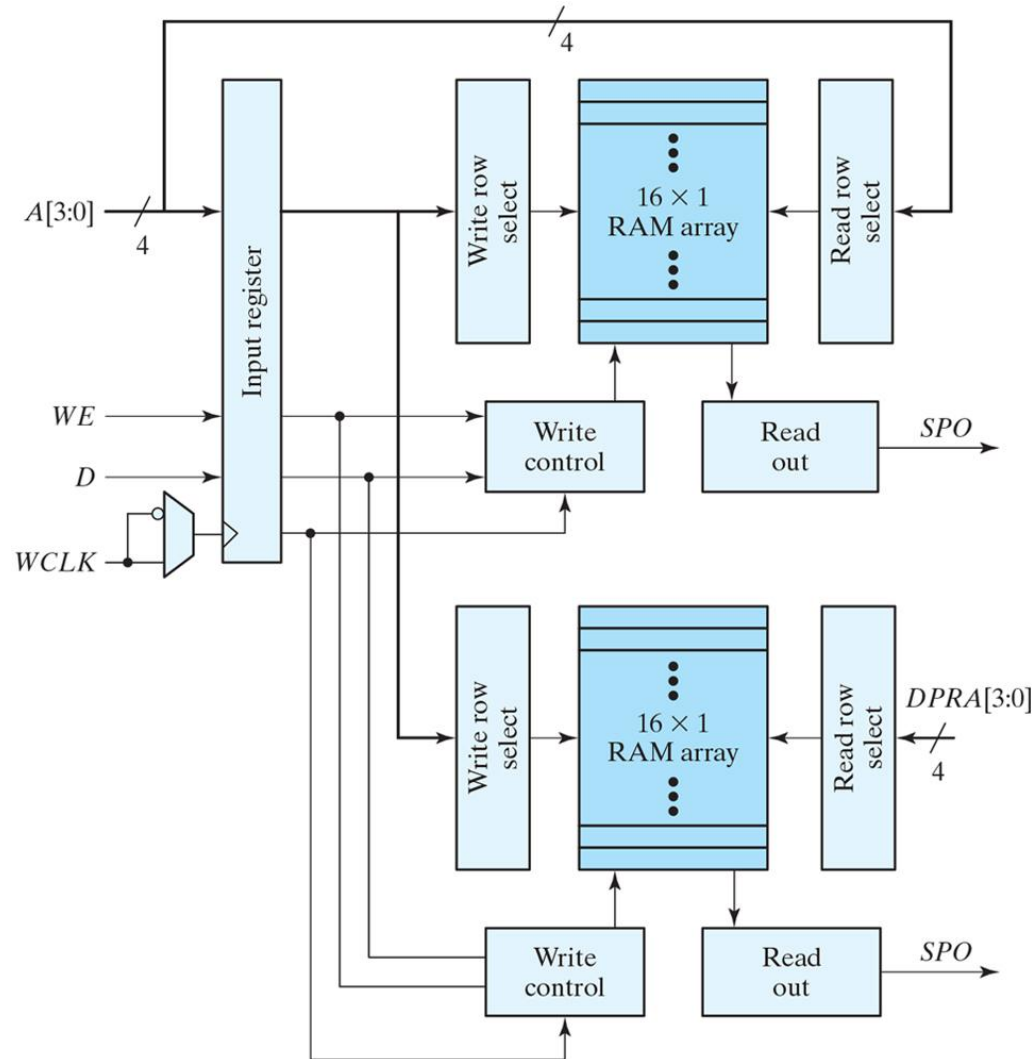


Figure 7.28
Spartan II architecture.

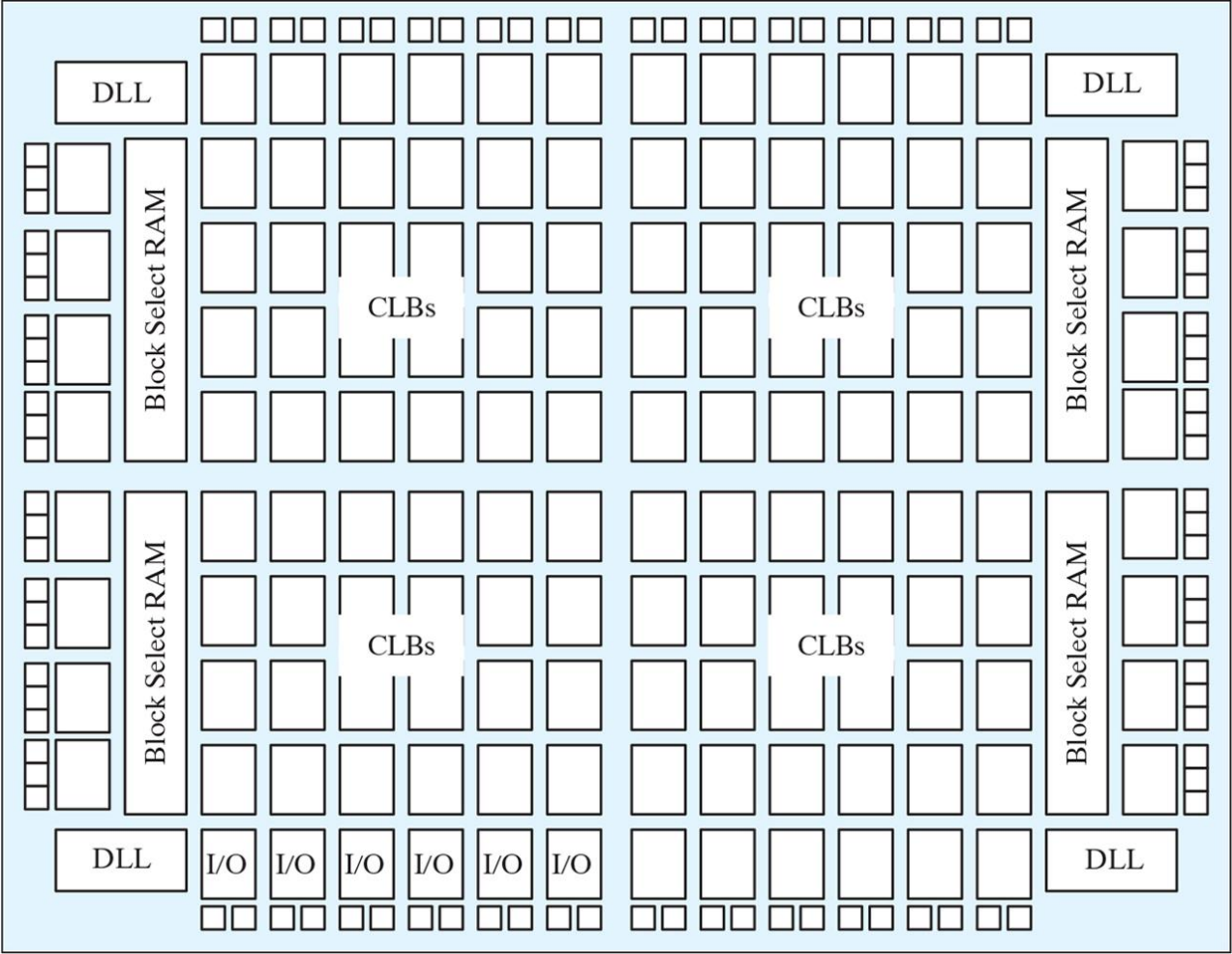


Figure 7.29
Spartan II CLB slice.

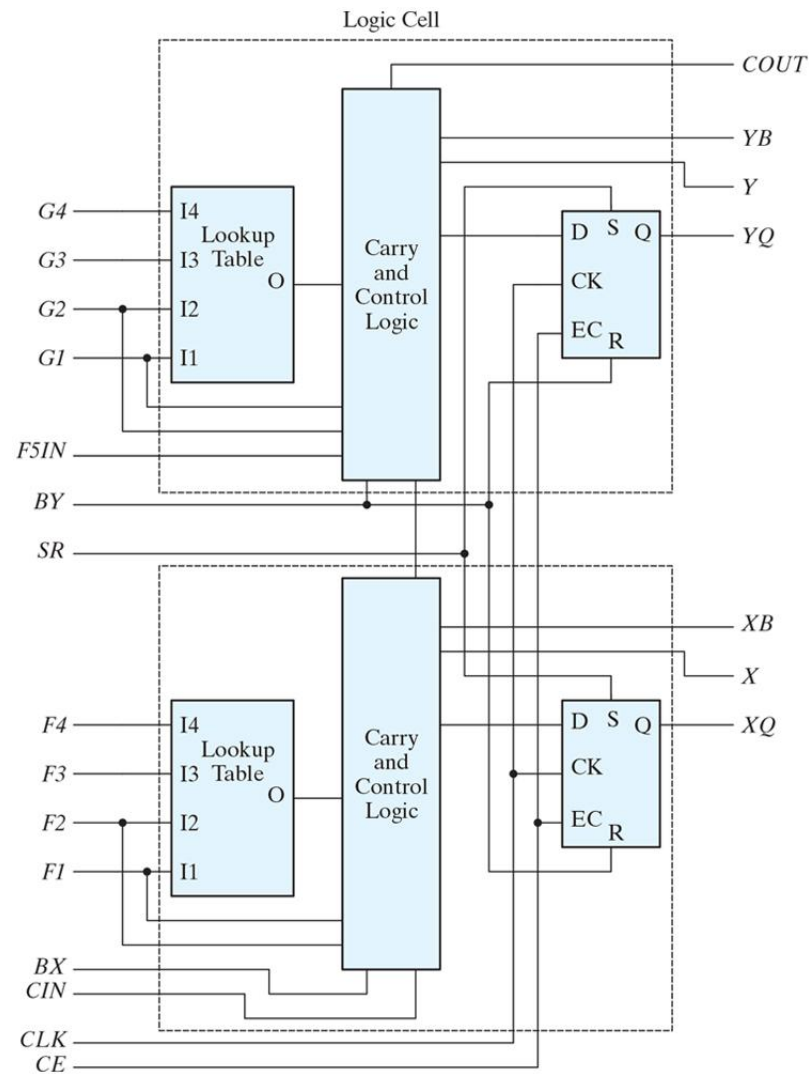


Figure 7.30
Spartan II IOB.

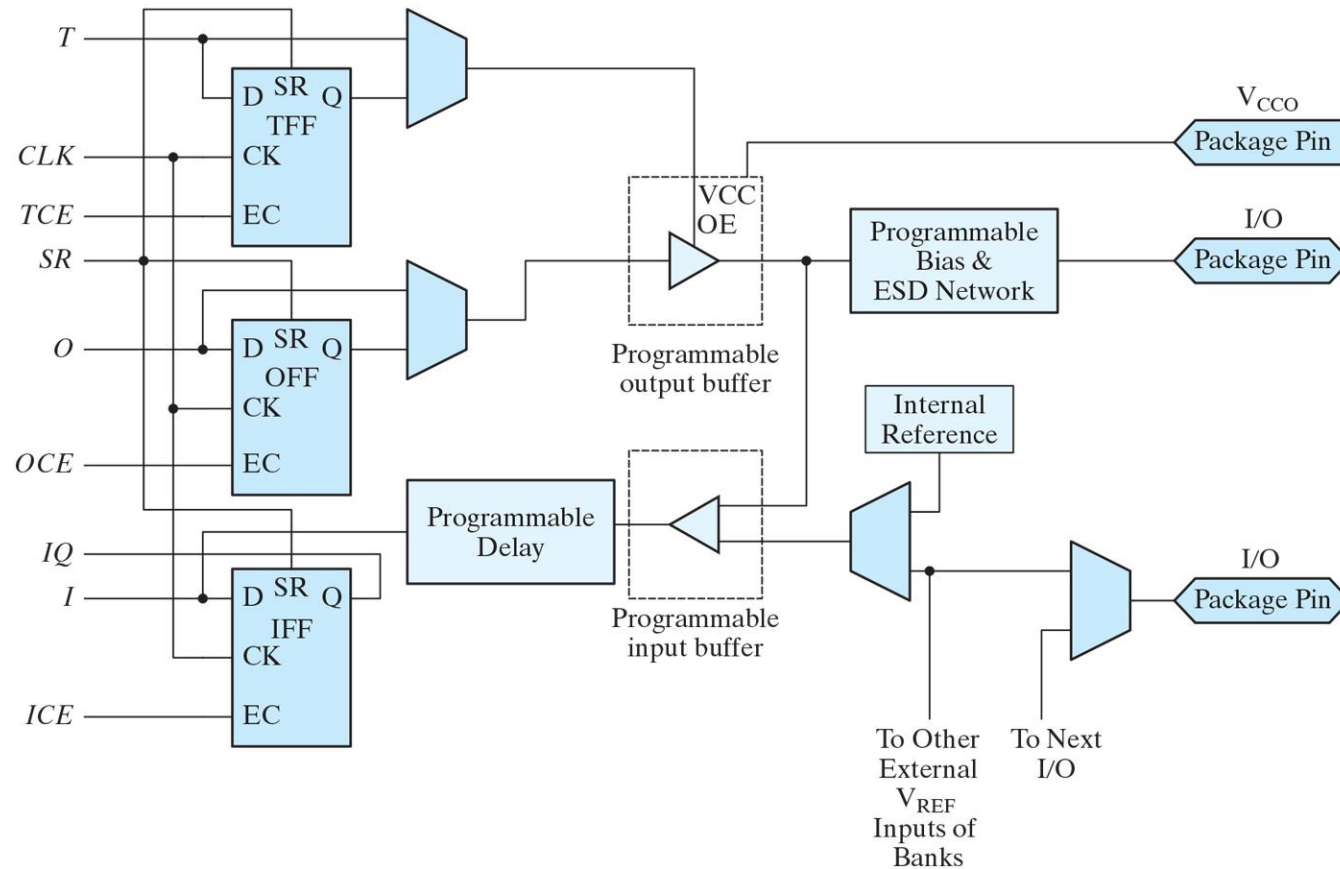


Table 7.7
Features of the Spartan-6 device family.

Device	Logic Cells	Configurable Logic Blocks (CLBs)				Block RAM Blocks		
		Slices	Flip-Flops	Max Distributed RAM(Kb)	DSPA1 Slices	18 Kb	Max (Kb)	CMTs
XC6SLX4	3,840	600	4,800	75	8	12	216	2
XC6SLX9	9,152	1,430	11,440	90	16	32	576	2
XC6SLX16	14,579	2,278	18,224	136	32	32	576	2
XC6SLX25	24,051	3,758	30,064	229	38	52	936	2
XC6SLX45	43,661	6,822	54,576	401	58	116	2,088	4
XC6SLX75	74,637	11,662	93,296	692	132	172	3,096	6
XC6SLX100	101,261	15,822	126,576	976	180	268	4,824	6
XC6SLX150	147,443	23,038	184,304	1,355	180	268	4,824	6
XC6SLX25T	24,051	3,758	30,064	229	38	52	936	2
XC6SLX45T	43,661	6,822	54,576	401	58	116	2,088	4
XC6SLX75T	74,637	11,662	93,296	692	132	172	3,096	6
XC6SLX100T	101,261	15,822	126,576	976	180	268	4,824	6
XC6SLX150T	147,443	23,038	184,304	1,355	180	268	4,824	6

Figure 7.31
Virtex overall architecture.

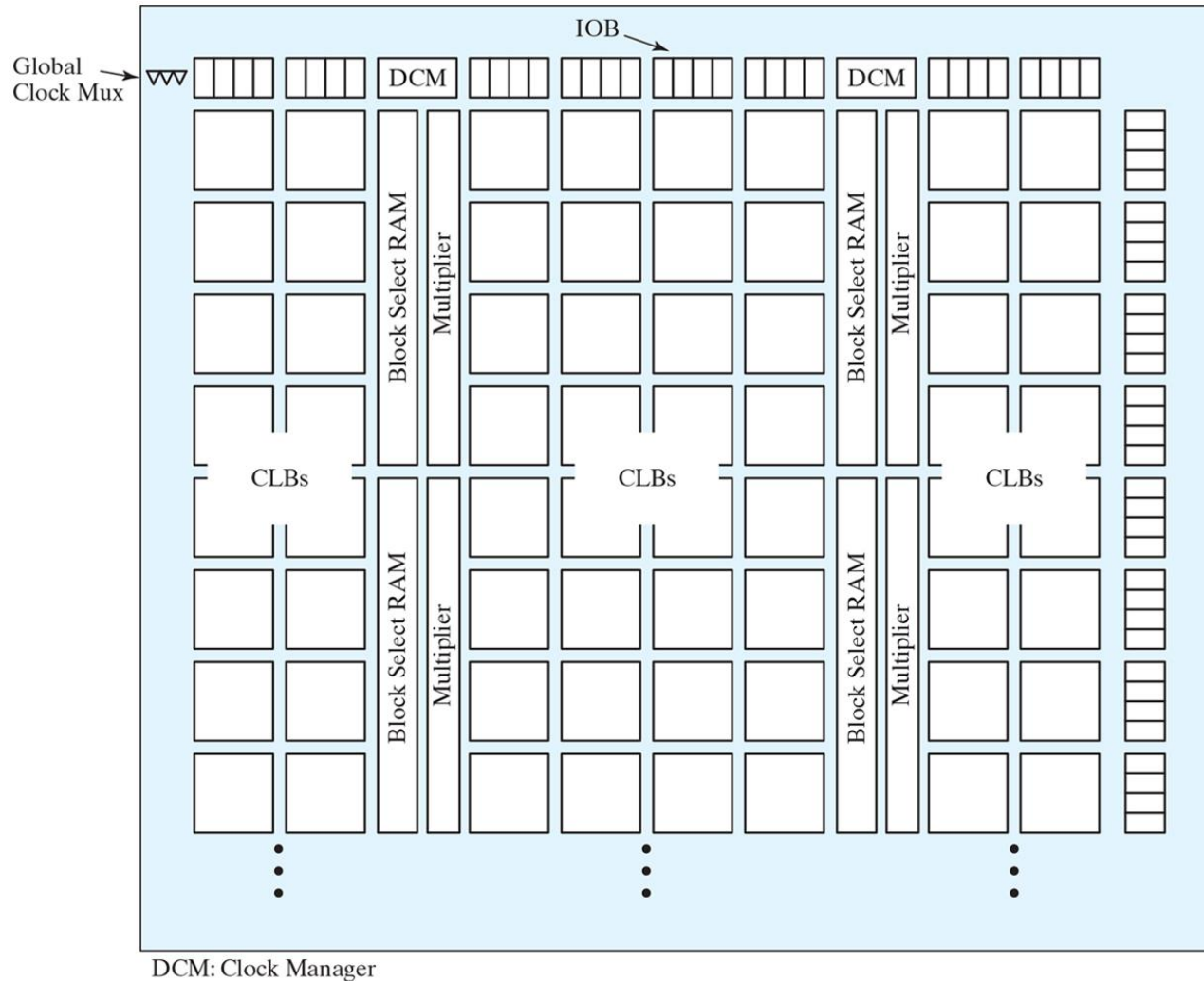


Figure 7.32
Virtex IOB block.

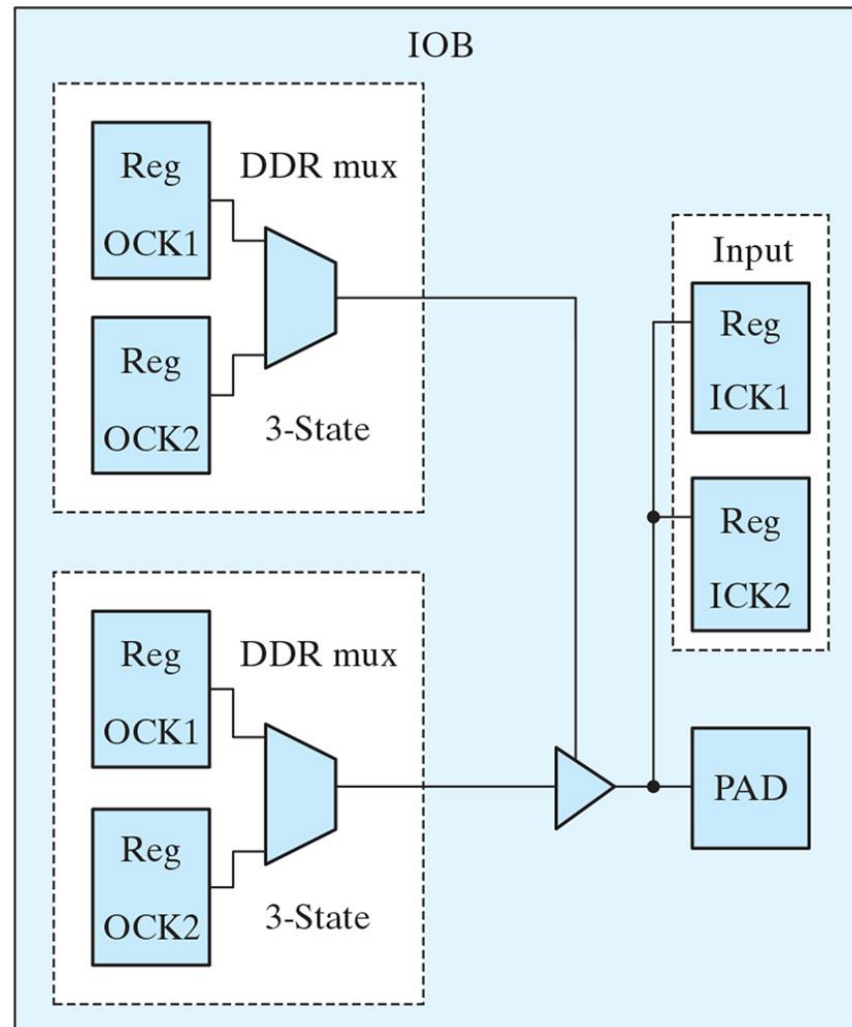
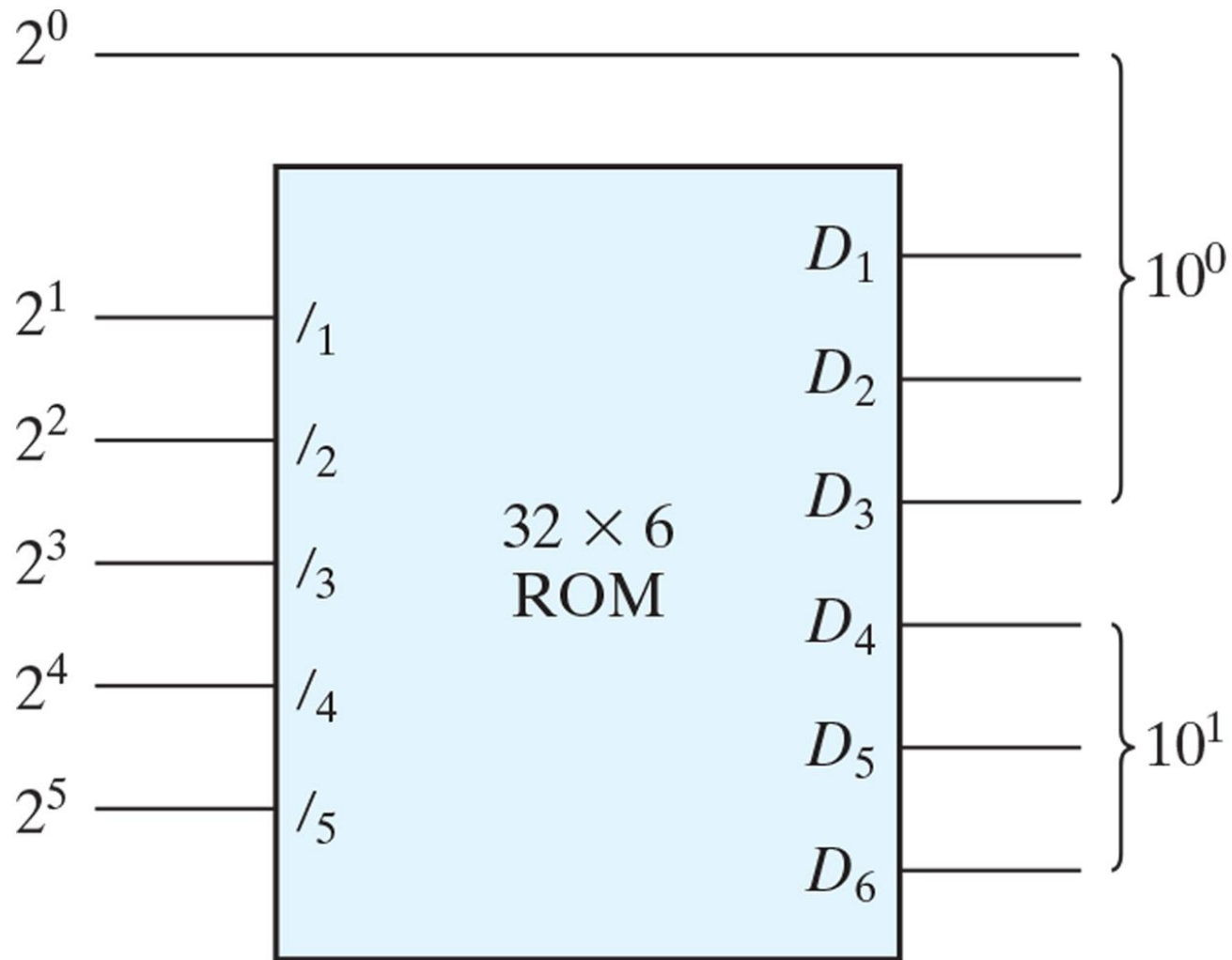


Table 7.8
Features of the Virtex Ultrascale device family.

Device	System Logic Cells (K)	CLB Flip-Flops (K)	CLB LUTs (K)	Max Distributed RAM (Mb)	Total Block RAM (Mb)	Clock Mgmt Tiles (CMTs)	DSP Slices
VU3P	862	788	394	12.0	25.3	10	2,280
VU5P	1,314	1,201	601	18.3	36.0	20	3,474
VU7P	1,724	1,576	788	24.1	50.6	20	4,560
VU9P	2,586	2,364	1,182	36.1	75.9	30	6,840
VU11P	2,835	2,592	1,296	36.2	70.9	12	2,088
VU13P	3,780	3,456	1,728	48.3	94.5	16	12,288

Figure P7.17



Problem 7.25

Inputs			Outputs			
<i>x</i>	<i>y</i>	<i>z</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
0	0	0	0	1	0	0
0	0	1	1	1	1	1
0	1	0	1	0	1	1
0	1	1	0	1	0	1
1	0	0	1	1	1	0
1	0	1	0	0	0	1
1	1	0	1	0	1	0
1	1	1	0	1	1	1